

AI/ML Intern — Technical & Business Assignment

Talknlock Pvt. Ltd. | AI & Marketing Intelligence

About the Assignment

Talknlock is a digital marketing company working across **Social Media Marketing, Performance Marketing, SEO, UGC, Video Production, and Content Marketing**.

We are now building our **AI/ML division from the ground up**.

This is not a conventional internship assignment. We are looking for someone who can potentially become one of the **founding members and future leaders of our AI function**.

We don't want someone who simply knows how to call an API or generate a chatbot.

We want someone who can look at a real business problem, understand the data behind it, determine whether AI/ML is actually required, build a practical solution, and explain how that solution can create measurable business value.

The Challenge

Build an AI-Powered Marketing Intelligence System

Imagine Talknlock manages **50–100 clients simultaneously**.

Each client generates large amounts of information:

- Social media performance
- Instagram engagement
- Content performance
- Ad campaign performance
- Website traffic
- Leads
- SEO performance
- Content calendars
- Client feedback
- Monthly reports
- Sales/conversion data

Currently, much of this information requires humans to manually analyse.

Your challenge is to design and build a **prototype AI/ML-powered Marketing Intelligence System** that can help a digital marketing agency make better decisions.

Part 1 — Business Problem Discovery

Before writing code, identify **3 major problems** that AI/ML could solve inside a digital marketing agency.

For each problem, explain:

1. What is the current problem?
2. Who experiences the problem?
3. What data would be required?
4. Why is AI/ML appropriate for solving it?
5. What would happen if the problem is not solved?
6. How would you measure the business impact?
7. What would a human currently have to do manually?

Important

We are specifically looking for candidates who understand that:

Not every problem requires AI.

If you believe a problem can be solved better using a simple rule-based system, automation, SQL, analytics, or traditional software, say so.

We will consider this a positive sign.

Part 2 — Marketing Dataset Analysis

Create or use a **synthetic dataset** representing at least:

500 marketing/content records

Each record may contain variables such as:

- Client
- Industry
- Platform
- Content type
- Content topic
- Posting day
- Posting time
- Reach
- Impressions
- Likes
- Comments
- Shares
- Saves

- Video views
- Watch time
- Clicks
- Leads
- Ad spend
- Revenue/conversion value

You are allowed to create realistic synthetic data if you don't have access to a real dataset.

Analyse the dataset

Identify:

- Which content performs best?
- Which content performs poorly?
- Which variables appear correlated with performance?
- Are there meaningful differences between industries?
- Are there meaningful differences between platforms?
- What anomalies or unusual patterns exist?
- What data would you want but don't have?

Your analysis should include appropriate **visualisations and statistical reasoning**.

Do not simply produce charts.

Explain what each important finding means for a marketing agency.

Part 3 — Build an ML Model

Using your dataset, build **at least one machine-learning model**.

You may choose one of the following problems:

Option A — Content Performance Prediction

Predict the expected performance of a piece of content before publishing.

Possible target:

- Engagement
- Reach
- Video views
- Leads
- Performance score

Option B — Lead Prediction

Predict whether a marketing lead is likely to convert.

Option C — Client Churn Prediction

Predict whether a client is at risk of leaving the agency.

Option D — Content Recommendation

Recommend what type of content should be created next based on historical performance.

Option E — Your Own Problem

You may propose another ML problem.

If you choose your own problem, explain why it is more valuable than the options above.

Part 4 — Model Evaluation

Do not stop at:

"My model achieved 92% accuracy."

Explain:

- Why you selected the model
- What features you used
- What features you excluded and why
- How you split the dataset
- What evaluation metric you selected
- Why that metric is appropriate
- Baseline performance
- Model performance
- Overfitting/underfitting considerations
- Limitations
- Potential data leakage
- What could go wrong in production

Where appropriate, compare **at least two approaches/models**.

We are more interested in your reasoning than the final accuracy number.

Part 5 — Explainability

A marketing manager who doesn't know machine learning should be able to understand your system.

For your model, answer:

"Why did the model make this prediction?"

Demonstrate how you would explain predictions to a non-technical employee.

You may use appropriate explainability techniques such as:

- Feature importance
- SHAP
- LIME
- Partial dependence
- Rule-based explanations

You do not need to use all of them.

Choose an appropriate approach and justify it.

Part 6 — Build a Simple Prototype

Create a basic working prototype/interface demonstrating how a Talknlock employee could use your system.

For example:

Input:

Industry → Fashion
Platform → Instagram
Content Type → Reel
Topic → Product Education
Posting Time → 7 PM

System Output:

Predicted performance score → 82/100

Expected engagement → X

Top factors influencing prediction:

1. Content format
2. Topic
3. Historical engagement
4. Posting time

Recommended action:

"Create more educational Reels around this topic."

The interface does not need to look beautiful.

We care about:

Functionality → Logic → Usefulness → Scalability

Part 7 — AI Layer

Now think beyond traditional ML.

Explain how you would combine your ML system with modern AI/LLM capabilities.

For example:

Marketing Data → ML Model → AI Reasoning Layer → Recommendation → Human Approval

Explain how an LLM could potentially:

- Summarise campaign performance
- Explain model predictions
- Generate recommendations
- Analyse client feedback
- Identify recurring client complaints
- Generate content insights
- Convert raw analytics into client-friendly reports

However, do **not** simply add an LLM because it is fashionable.

Explain:

What should be handled by traditional software/ML and what should be handled by an LLM?

Part 8 — Production Architecture

Assume Talklock wants to turn your prototype into a real internal product.

Design a high-level architecture covering:

- Data collection
- Data storage
- Data processing
- ML model
- AI/LLM layer
- API/backend
- Dashboard
- Authentication
- Monitoring

- Model retraining
- Data privacy
- Security

You may present this as an architecture diagram.

We are not expecting enterprise-level architecture from an intern.

We are evaluating whether you can **think beyond the notebook**.

Part 9 — Business Case

Imagine your system is successfully implemented across Talknlock.

Estimate:

Potential impact

For example:

- Hours saved per month
- Reduction in manual reporting
- Improvement in content performance
- Improvement in lead qualification
- Reduction in client churn
- Additional revenue opportunities

You don't need to have exact numbers.

Make reasonable assumptions and clearly state them.

Then answer:

If you were the CEO, would you invest money in building this product? Why or why not?

Part 10 — Future Vision

This is the most important part.

Assume you join Talknlock and are given **12 months to build the AI/ML department from zero**.

Create a roadmap covering:

Month 1–3

What would you build first?

Month 4–6

What would you build next?

Month 7–9

What systems/products would you introduce?

Month 10–12

How would you turn the AI function into a competitive advantage for Talknlock?

Then answer:

What could Talknlock's AI division look like 3 years from now?

Think beyond internal automation.

Consider potential:

- AI marketing products
- SaaS products
- AI-powered reporting
- Marketing intelligence platforms
- Predictive analytics
- AI agents
- Client-facing products
- Proprietary datasets
- AI-powered campaign optimisation

Deliverables

Submit the following:

1. Source Code

GitHub repository containing:

- Clean code
- README
- Requirements/dependencies
- Dataset generation/loading
- Data processing
- Model training

- Evaluation
- Prototype/application

2. Technical Report

Maximum **15 pages** covering:

- Problem definition
- Data
- Exploratory analysis
- Model
- Evaluation
- Explainability
- AI layer
- Architecture
- Limitations
- Business impact
- Future roadmap

3. Architecture Diagram

One clear diagram explaining the proposed production system.

4. Working Prototype

A basic functional demo.

5. 5-Minute Video

Record yourself explaining:

"What I built, why I built it, what I learned, and what I would build next."

You must appear on camera and explain the project yourself.

STRICT NO-AI POLICY

This assignment must be completed independently.

You are **NOT permitted to use generative AI tools** to complete this assignment.

This includes, but is not limited to:

- ChatGPT
- Claude
- Gemini
- Perplexity
- GitHub Copilot
- Cursor AI

- Replit AI
- AI coding assistants
- AI-generated reports
- AI-generated presentations
- AI-generated diagrams
- Any other generative AI tool

You may use:

- Official documentation
- Research papers
- Stack Overflow
- Kaggle
- Python documentation
- Library documentation
- Conventional search engines
- Books and educational resources

You must understand and be able to explain **every significant part of your submission**.

Important

We are not trying to test whether you can produce a perfect answer.

We are testing:

Your thinking.

Your technical fundamentals.

Your curiosity.

Your problem-solving ability.

Your ability to learn.

Your ability to build.

A simple project with strong reasoning is significantly better than an impressive-looking project that you cannot explain.

AI Verification Round

Shortlisted candidates will participate in a **30–45 minute technical discussion**.

During this discussion, we may ask you to:

- Explain your code
- Modify part of your model
- Explain why you selected a particular algorithm
- Debug a section of your code
- Reproduce part of your analysis
- Explain your dataset
- Defend your assumptions

- Explain your architecture
- Propose improvements
- Solve a new marketing problem without preparation

You may also be given a **live problem unrelated to your submitted solution**.

If you cannot explain your own submission, the assignment will be considered invalid.

Evaluation Criteria

Category	Weight
Problem-solving & business thinking	20%
Python / programming fundamentals	15%
Data analysis & statistics	15%
Machine learning fundamentals	15%
Model evaluation & reasoning	10%
AI/LLM understanding	10%
System/architecture thinking	5%
Communication	5%
Innovation & initiative	5%
Total	100%

What We Are Looking For

The ideal candidate does **not** necessarily have the most complicated model.

We are looking for someone who:

- Is genuinely curious about AI/ML
- Understands fundamentals
- Can work with imperfect data
- Thinks about business outcomes
- Questions assumptions
- Can explain technical concepts simply
- Is comfortable learning independently
- Can build rather than only theorise
- Thinks about scalability
- Has the ambition to eventually own a function

Bonus

Candidates who identify a significantly better AI/ML opportunity for Talknlock than the one described in this assignment are encouraged to propose it.

You are not being evaluated on whether you follow our suggested solution.

We want to see how you think.

Time Limit

Recommended effort: 10–14 hours

Submission deadline will be communicated separately.

We deliberately recommend a limited time commitment.

Do not spend 50 hours trying to make the project perfect.

Prioritise:

Good thinking > complicated technology

Business value > unnecessary features

Understanding > presentation

Final Question

At the end of your submission, answer this in **500 words or less**:

"If Talknlock gives you the opportunity to build its AI/ML department from scratch, why should we trust you with that responsibility?"

Be honest.

We are not looking for the most impressive answer.

We are looking for the person behind it.

Internal Hiring Note

This assignment is designed to identify a potential **founding AI/ML team member**, not simply an intern.

The strongest candidates should demonstrate the ability to progress from:

Data Analyst → ML Builder → AI Engineer → Product Thinker → AI/ML Department Leader

A candidate who produces a technically modest solution but demonstrates exceptional **ownership, curiosity, reasoning, and business understanding** should be considered more seriously than a candidate who produces an advanced project they cannot defend.